

Increased hair testosterone but unaltered hair cortisol in female patients with borderline personality disorder.

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A number of studies have reported on dysfunctions in steroid secretion, including altered cortisol and testosterone levels in borderline personality disorder (BDP) patients compared to healthy controls. The present study extends findings from blood and saliva studies to the cumulative measure of hair steroids. We investigated women with BPD (n=18) and age- and education-matched healthy women (n=17). We did not find differences between BPD patients and healthy women (p=0.40) concerning hair cortisol levels but increased hair testosterone levels among BPD patients compared to controls (p=0.03). These results remained when restricting the analyses to unmedicated patients. Our data indicate altered long-term testosterone but not cortisol levels in females with BPD. Future studies should address the possible impact of altered testosterone on medical illness processes including metabolic syndrome in this population.

X Chromosome Factor Kdm6a Enhances Cognition Independent of Its Demethylase Function in the Aging XY Male Brain.

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Males exhibit shorter life span and more cognitive deficits, in the absence of dementia, in aging human populations. In mammals, the X chromosome is enriched for neural genes and is a major source of biologic sex difference, in part, because males show decreased expression of select X factors (XY). While each sex (XX and XY) harbors one active X due to X chromosome inactivation in females, some genes, such as Kdm6a, transcriptionally escape silencing in females-resulting in lower transcript levels in males. Kdm6a is a known histone demethylase (H3K27me2/3) with multiple functional domains that is linked with synaptic plasticity and cognition. Whether elevating Kdm6a could benefit the aged male brain and whether this requires its demethylase function remains unknown. We used lentiviral-mediated overexpression of the X factor in the hippocampus of aging male mice and tested their cognition and behavior in the Morris water-maze. We found that acutely increasing Kdm6a-in a form without demethylase function-selectively improved learning and memory, in the aging XY brain, without altering total activity or anxiety-like measures. Further understanding the demethylase-independent downstream mechanisms of Kdm6a may lead to novel therapies for treating age-induced cognitive deficits in both sexes.

Evidence from Turner's syndrome of an imprinted X-linked locus affecting cognitive function.

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Turner's syndrome is a sporadic disorder of human females in which all or part of one X chromosome is deleted. Intelligence is usually normal but social adjustment problems are common. Here we report a study of 80 females with Turner's syndrome and a single X chromosome, in 55 of which the X was maternally derived (45,X[m]) and in 25 it was of paternal origin (45,X[p]). Members of the 45,X[p] group were significantly better adjusted, with superior verbal and higher-order executive function skills, which

mediate social interactions. Our observations suggest that there is a genetic locus for social cognition, which is imprinted and is not expressed from the maternally derived X chromosome. Neuropsychological and molecular investigations of eight females with partial deletions of the short arm of the X chromosome indicate that the putative imprinted locus escapes X-inactivation, and probably lies on Xq or close to the centromere on Xp. If expressed only from the X chromosome of paternal origin, the existence of this locus could explain why 46,XY males (whose single X chromosome is maternal) are more vulnerable to developmental disorders of language and social cognition, such as autism, than are 46,XX females.

Cognitive, behavioral, and neural consequences of sex chromosome aneuploidy.

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The X chromosome has played a critical role in the development of sexually selected characteristics for over 300 million years, and during that time it has accumulated a disproportionate number of genes concerned with mental functions. There are relatively specific effects of X-linked genes on social cognition, language, emotional regulation, visuospatial, and numerical skills. Many human X-linked genes outside the X-Y pairing pseudoautosomal regions escape X-inactivation. Dosage differences in the expression of such genes (which constitute at least 15% of the total) are likely to play an important role in male-female neural differentiation, and in cognitive deficits and behavioral characteristics, particularly in the realm of social communication, that are associated with sex chromosome aneuploidies. © 2016 Wiley Periodicals, Inc.

[Monozygous female twins discordant for a pattern of X chromosome inactivation as a model for study on X-linked intellectual traits].

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Ring the last decades, numerous genes for general cognitive ability were identified on human X-chromosome. They were discovered primarily because of X-linked mutations causing nonspecific mental retardation in males. Evidence for imprinted loci on the X chromosome affecting neurodevelopment was found in studies on 45,X females. Investigation of transmission of X-linked traits in normal individuals might further contribute to problem of shaping human being's mind ability. We suggest monozygous female twins discordant for a parent-of-origin of the X chromosome inactivation to be a proper subject for such explorations.

Skewed X-chromosome inactivation is a common feature of X-linked mental retardation disorders.

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Some deleterious X-linked mutations may result in a growth disadvantage for those cells in which the mutation, when on the active X chromosome, affects cell proliferation or viability. To explore the relationship between skewed X-chromosome inactivation and X-linked mental retardation (XLMR)

disorders, we used the androgen receptor X-inactivation assay to determine X-inactivation patterns in 155 female subjects from 24 families segregating 20 distinct XLMR disorders. Among XLMR carriers, approximately 50% demonstrate markedly skewed X inactivation (i.e., patterns $>$ or $=80:20$), compared with only approximately 10% of female control subjects (P or $=80:20$). The XLMR mutations are present on the preferentially inactive X chromosome in all 20 informative female subjects from these families, indicating that skewing is due to selection against those cells in which the XLMR mutation is on the active X chromosome.

X chromosome inactivation and X-linked mental retardation.

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The expression of X-linked genes in females heterozygous for X-linked defects can be modulated by epigenetic control mechanisms that constitute the X chromosome inactivation pathway. At least four different effects have been found to influence, in females, the phenotypic expression of genes responsible for X-linked mental retardation (XLMR). First, non-random X inactivation, due either to stochastic or genetic factors, can result in tissues in which one cell type (for example, that in which the X chromosome carrying a mutant XLMR gene is active) dominates, instead of the normal mosaic cell population expected as a result of random X inactivation. Second, skewed inactivation of the normal X in individuals carrying a deletion of part of the X chromosome has been documented in a number of mentally retarded females. Third, functional disomy of X-linked genes that are expressed inappropriately due to the absence of X inactivation has been found in mentally retarded females with structurally abnormal X chromosomes that do not contain the X inactivation center. And fourth, dose-dependent overexpression of X-linked genes that normally "escape" X inactivation may account for the mental and developmental delay associated with increasing numbers of otherwise inactive X chromosomes in individuals with X chromosome aneuploidy.

X-linked genes and mental functioning.

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The X-chromosome has played a crucial role in the development of sexually selected characteristics for over 300 million years. During that time it has accumulated a disproportionate number of genes concerned with mental functions. Evidence is emerging, from studies of both humans and mice, for a general influence upon intelligence (as indicated by the large number of X-linked mental retardation syndromes). In addition, there is evidence for relatively specific effects of X-linked genes on social-cognition and emotional regulation. Sexually dimorphic processes could be influenced by several mechanisms. First, a small number of X-linked genes are apparently expressed differently in male and female brains in mouse models. Secondly, many human X-linked genes outside the X-Y pairing pseudoautosomal regions escape X-inactivation. Dosage differences in the expression of such genes (which might comprise at least 20% of the total) are likely to play an important role in male-female neural differentiation. To date, little is known about the process but clues can be gleaned from the study of X-monosomic females who are haploinsufficient for expression of all non-inactivated genes relative to 46,XX females. Finally, from studies of both X-monosomic humans (45,X) and mice (39,X), we are learning more about the influences of X-linked imprinted genes upon brain structure and function. Surprising specificity of effects has been described in both species, and identification of candidate genes

cannot now be far off.

Intelligence Quotient Variability in Klinefelter Syndrome Is Associated With GTPBP6 Expression Under Regulation of X-Chromosome Inactivation Pattern.

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Klinefelter syndrome (KS) displays a broad dysmorphological, endocrinological, and neuropsychological clinical spectrum. We hypothesized that the neurocognitive dysfunction present in KS relies on an imbalance in X-chromosome gene expression. Thus, the X-chromosome inactivation (XCI) pattern and neurocognitive X-linked gene expression were tested and correlated with intelligence quotient (IQ) scores. We evaluated 11 KS patients by (a) IQ assessment, (b) analyzing the XCI patterns using both HUMARA and ZDHHC15 gene assays, and (c) blood RT-qPCR to investigate seven X-linked genes related to neurocognitive development (GTPBP6, EIF2S3, ITM2A, HUWE1, KDM5C, GDI1, and VAMP7) and XIST in comparison with 14 (male and female) controls. Considering IQ 80 as the standard minimum reference, we verified that the variability in IQ scores in KS patients seemed to be associated with the XCI pattern. Seven individuals in the KS group presented a random X-inactivation (RXI) and lower average IQ than the four individuals who presented a skewed X-inactivation (SXI) pattern. The evaluation of gene expression showed higher GTPBP6 expression in KS patients with RXI than in controls ($p = 0.0059$). Interestingly, the expression of GTPBP6 in KS patients with SXI did not differ from that observed in controls. Therefore, our data suggest for the first time that GTPBP6 expression is negatively associated with full-scale IQ under the regulation of the type of XCI pattern. The SXI pattern may regulate GTPBP6 expression, thereby dampening the impairment in cognitive performance and playing a role in intelligence variability in individuals with KS, which warrants further mechanistic investigations.

Ring-X chromosomes: their cognitive and behavioural phenotype.

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We tested the cognitive abilities and educational attainments of 47 patients with a ring X chromosome, to evaluate the extent to which these variables correlated with failure of r(X) inactivation and with mosaicism. We found possession of a r(X) chromosome was associated with an increased risk of significant learning difficulties, and with associated behavioural maladjustment, compared with 45,X Turner females. Nearly a third had been educated outside mainstream schools. The proportion of cells in peripheral blood containing an inactivated r(X) chromosome was negatively correlated with nonverbal IQ. The parental origin of the normal chromosome did not appear to affect adjustment or abilities. In a minority of r(X) cases associated with mental retardation, there had been a failure to inactivate the ring, due to loss of the XIST locus. However, failure of X-inactivation was not necessarily associated with a severe phenotype. The degree of impairment in IQ depended on the size of the active ring, and hence was proportionate to the number of (as yet unidentified) genes whose functional disomy affected brain development and functioning.